

Assignment

Fourier Transforms

1. Find the Fourier transform of $f(x) = e^{-a^2 x^2}$, where a is a real constant.
2. Find the Fourier transform of $f(x) = e^{-a|x|}$, where $a > 0$ is a real constant.
3. Find the Fourier transform of $f(x) = x e^{-a|x|}$, where $a > 0$ is a real constant.
4. Find the Fourier transform of the following functions:
 (i) $f(x) = \begin{cases} \sin x & \text{for } 0 < x < \pi \\ 0, & \text{otherwise} \end{cases}$ (ii) $f(x) = \begin{cases} \cos x & \text{for } 0 < x < \pi \\ 0, & \text{otherwise} \end{cases}$
5. Find the Fourier transform of the following functions:
 (i) $f(x) = \begin{cases} x & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$ (ii) $f(x) = \begin{cases} x^2 & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$
6. Find the complex Fourier transform of the function $f(x) = \begin{cases} 1 & \text{for } |x| \leq a \\ 0 & \text{for } |x| > a \end{cases}$.
 Hence evaluate $\int_0^\infty \frac{\sin x}{x} dx$.
7. Find the complex Fourier transform of the function $f(x) = \begin{cases} 1 - |x| & \text{for } |x| \leq 1 \\ 0 & \text{for } |x| > 1 \end{cases}$. Hence deduce that $\int_0^\infty \frac{\sin^2 x}{x^2} dx = \frac{\pi}{2}$.
8. Find the complex Fourier transform of the function $f(x) = \begin{cases} 1 - x^2 & \text{for } |x| \leq 1 \\ 0 & \text{for } |x| > 1 \end{cases}$.
 Hence evaluate $\int_0^\infty \left(\frac{x \cos x - \sin x}{x^3} \right) \cos \frac{x}{2} dx$.
9. Find the complex Fourier transform of the function $f(x) = \begin{cases} 1 & \text{for } |x| < a \\ 0 & \text{for } |x| > a \end{cases}$.
 Hence evaluate (i) $\int_{-\infty}^\infty \left(\frac{\sin ap \cos px}{p} \right) dp$ (ii) $\int_0^\infty \left(\frac{\sin p}{p} \right) dp$.
10. Find the complex Fourier transform of the functions
 (i) $f(x) = \begin{cases} 1 + \left(\frac{x}{a}\right) & \text{for } -a < x < 0 \\ 1 - \left(\frac{x}{a}\right) & \text{for } 0 < x < a \\ 0 & \text{otherwise} \end{cases}$ (ii) $f(x) = \begin{cases} e^x, & |x| < a \\ 0, & \text{otherwise} \end{cases}$